

ALFIDO TECH MACHINE LEARNING INTERNSHIP



PROJECT REPORT

Task Name: Iris Classification

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1.INTRODUCTION:

Machine learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions.

2.OBJECTIVE:

The objective of this project is to classify iris flowers using machine learning algorithms.

3.Tools & Technologies Method:

Python

JupyterNotebook

Pandas

Numpy

Matplotlib

Seaborn

Scikit-learn

4.Dataset Information:

The Iris dataset contains flower measurements such as sepal -length,sepal width, petal length, petal width.

5.METHODOLOGY:

1.Imported required libraries

2.Loaded Dataset

3.Performed Data analysis

4.Created graphs

5.Split Dataset

6.Trained model

7.Predicted output

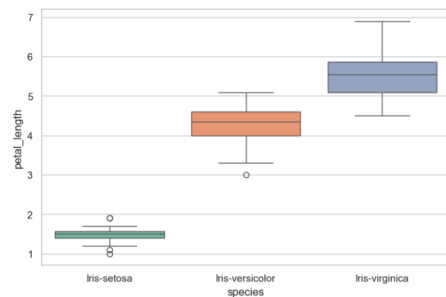
8.Evaluated accuracy

6.DATA VISUALIZATION:

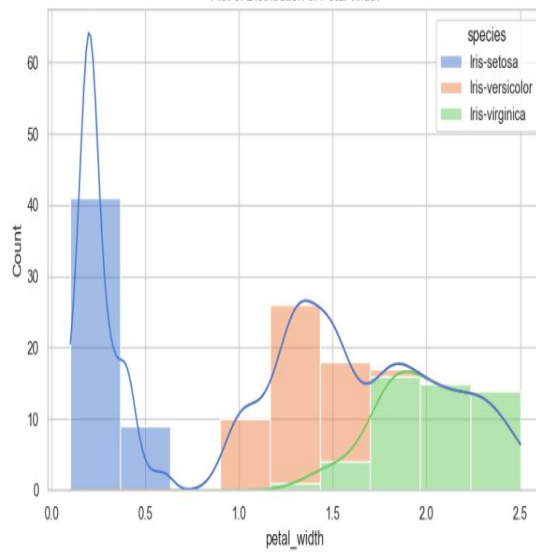

```

best_model_instance = model
print("\n\n>> Selected Best Model: (best_model_name) ((best_accuracy*100:.2f)% Accuracy)")
X_te_final = X_test_scaled if best_model_name != "Decision Tree" else X_test
final_preds = best_model_instance.predict(X_te_final)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, final_preds))
print("\nClassification Report:")
print(classification_report(y_test, final_preds))
joblib.dump(best_model_instance, 'best_iris_model.pkl')
print("Best model saved successfully!")

```



Plot 3: Distribution of Petal Width



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7.MODEL TRAINING:

Logistic Regression algorithm, K-NearestNeighbor, Decision Tree was used to train the model.

8.PREDICTION RESULTS:

The model successfully predicted flower categories.

9.ACCURACY:

The model achieved 93.33% accuracy



10.CONCLUSION:

Successfully completed iris classification using machine learning techniques.